

Energy Transfer

Answer sheet · 25 marks total

Section A — Key ideas

A1. Mechanically (1) electrically (1) by heating (1) by radiation (1)

A2. Energy cannot be created or destroyed, only transferred, stored or dissipated (1).

Section B — Kinetic & gravitational potential energy

B1. $E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 1200 \times 15^2$ (1) $E_k = 135,000 \text{ J}$ ($1.35 \times 10^5 \text{ J}$) (1)

B2. $E_p = mgh = 0.4 \times 9.8 \times 3$ (1) $E_p = 11.76 \text{ J}$ (1)

Section C — Elastic potential energy & specific heat capacity

C1. $E_e = \frac{1}{2}ke^2$ (1) $E_e = \frac{1}{2} \times 25 \times 0.08^2$ (1) $E_e = 0.08 \text{ J}$ (1)

C2. $\Delta\theta = \Delta E / (mc)$ (1) $\Delta\theta = 42000 / (2 \times 4200)$ (1) $\Delta\theta = 5^\circ\text{C}$ (1)

Section D — Power & efficiency

D1. 3 minutes = 180 s (1) $E = Pt = 2000 \times 180$ (1) $E = 360,000 \text{ J}$ ($3.6 \times 10^5 \text{ J}$) (1)

D2. efficiency = $12 / 60$ (1) efficiency = $0.2 = 20\%$ (1)

Section E — Extended reasoning

E1. Some of the input energy is always dissipated to the surroundings, usually as heat, due to friction/resistance (1). This wasted energy cannot be usefully recovered, so useful output energy transfer is always less than total input energy transfer (1). Lubrication (oiling/greasing moving parts) reduces friction and unwanted heat transfer (1) — insulation also accepted for a heat-transfer context.

Section F — Energy stores

(a) Elastic potential (1) (b) Chemical (1)

Marking note: award method marks for a correctly stated equation and correct substitution even if the final answer contains an arithmetic error. Accept answers in standard form where shown as equivalent.